

Global Health System Performance Diagnostic

Findings Briefing | September 2026 | Ilias Kubica

Situation

Health spending is often treated as a shorthand for how strong a health system is, on the assumption that more money spent leads to proportionally better outcomes. This diagnostic tests that assumption directly across 16 countries, using life expectancy as the outcome measure and three independent indicators as predictors: health expenditure, physician density, and maternal mortality. Where actual outcomes diverge from what an indicator alone would predict, that gap is treated as a signal worth investigating further.

Finding 1: Workforce capacity predicts outcomes more reliably than spend

Physician density correlates with life expectancy more strongly ($r=0.82$) than health expenditure does ($r=0.59$), shown in the *Physician Density vs Life Expectancy* and *Health Spend vs Life Expectancy* visuals. This suggests that how many trained healthcare workers a population actually has access to may be a more reliable indicator of system strength than total spend, which can be absorbed by cost inflation, inefficiency, or non-frontline allocation without ever reaching outcomes.

Finding 2: Nigeria's outcome gap is metric-specific, not systemic

Nigeria underperforms sharply on both spend (16.1 years below what would be predicted, the largest gap in the dataset) and physician density (10.8 years below prediction), clearly visible in both bar charts. On maternal mortality specifically, though, Nigeria actually overperforms its predicted life expectancy by 4.1 years, shown in the *Maternal Mortality* table. That's a meaningful distinction. It suggests Nigeria's life expectancy gap isn't evenly explained across every input, but is disproportionately tied to workforce and financial capacity specifically, rather than a uniform failure across the whole health system.

Finding 3: The US is a genuine efficiency outlier, not just a high spender

The US spends more per capita than any other country in the dataset by a wide margin, nearly double the next highest, yet underperforms its predicted life expectancy by 9.2 years. This is visible on both the bar chart and the scatter plot, where it sits far to the right on spend but below several lower spending countries in outcomes. Meanwhile, Bangladesh, Vietnam, Mexico, Brazil, and India all cluster on the low end of spend while outperforming expectations, Bangladesh most notably (up 4.1 years on spend, up 7.3 on physician density). Efficient outcomes are clearly achievable well below the US's spending level.

Recommendation

Resource allocation should weight workforce capacity and outcome specific interventions, like maternal health programmes, above undifferentiated spend increases. The data suggests that where and how resources are deployed drives outcomes more than the total amount spent, a finding directly relevant to any health system deciding whether a bigger budget alone will meaningfully improve population health.

Methodology and limitations: Based on World Bank Open Data, 2023 or the latest available year per indicator, across 16 countries. Physician density data was only available for 8 of the 16 countries in 2023, which limits the scope of that specific analysis. These are correlational findings. They identify association, not causation, and the residuals highlight where further investigation is warranted, not confirmed root causes.

Full analysis, code, and interactive dashboard: github.com/iliaskubica/global-health-diagnostic